

An isolated point in a countable Baire space

MathlibAnnex.Topology.exists_isOpen_singleton

theorem

Applies the Baire property to the closed cover by singletons.

Statement

Let X be a nonempty countable topological space which is T_1 and Baire. Then some point $x \in X$ is isolated: its singleton $\{x\}$ is open.

Assumptions

The T_1 condition says that singletons are closed. No metric, completeness, compactness or Hausdorff assumption beyond T_1 is required.

Conclusion

At least one singleton is open; the conclusion does not say that every point is isolated.

Proof route

A countable closed cover of a nonempty Baire space has a member with nonempty interior.

▼ Proof steps

1. The T_1 property makes each $F_x = \{x\}$ closed. Since X is countable,

$$X = \bigcup_{x \in X} F_x$$

is a countable closed cover. The Closed-cover Baire theorem with the singleton family applies because X is nonempty and Baire.

2. It yields x with $\text{int}(F_x) \neq \emptyset$. Choose y in that interior. Since $\text{int}(F_x) \subseteq \{x\}$, $y = x$, so $x \in \text{int}\{x\}$. Thus

$$\{x\} \subseteq \text{int}\{x\} \subseteq \{x\},$$

and $\{x\}$ equals its interior and is open.

Main citations

- Exact theorem and proof

Lean source signature (exact)

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theorem exists_isOpen_singleton [Nonempty X] [Countable X] [T1Space X]
  [BaireSpace X] :  $\exists x : X, \text{IsOpen } (\{x\} : \text{Set } X)$ 
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In the source	Mathematical meaning
[Nonempty X]	The topological space X has at least one point, needed for the nonempty closed-cover conclusion.
[Countable X]	The cover indexed by the points of X is countable.
[T1Space X]	Every singleton $\{x\}$ is closed.
[BaireSpace X]	X has the Baire property, so a countable closed cover of nonempty X has a member with nonempty interior.
$\exists x : X, \text{IsOpen } (\{x\} : \text{Set } X)$	There exists a point whose one-point subset is open in the original topology of X .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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